



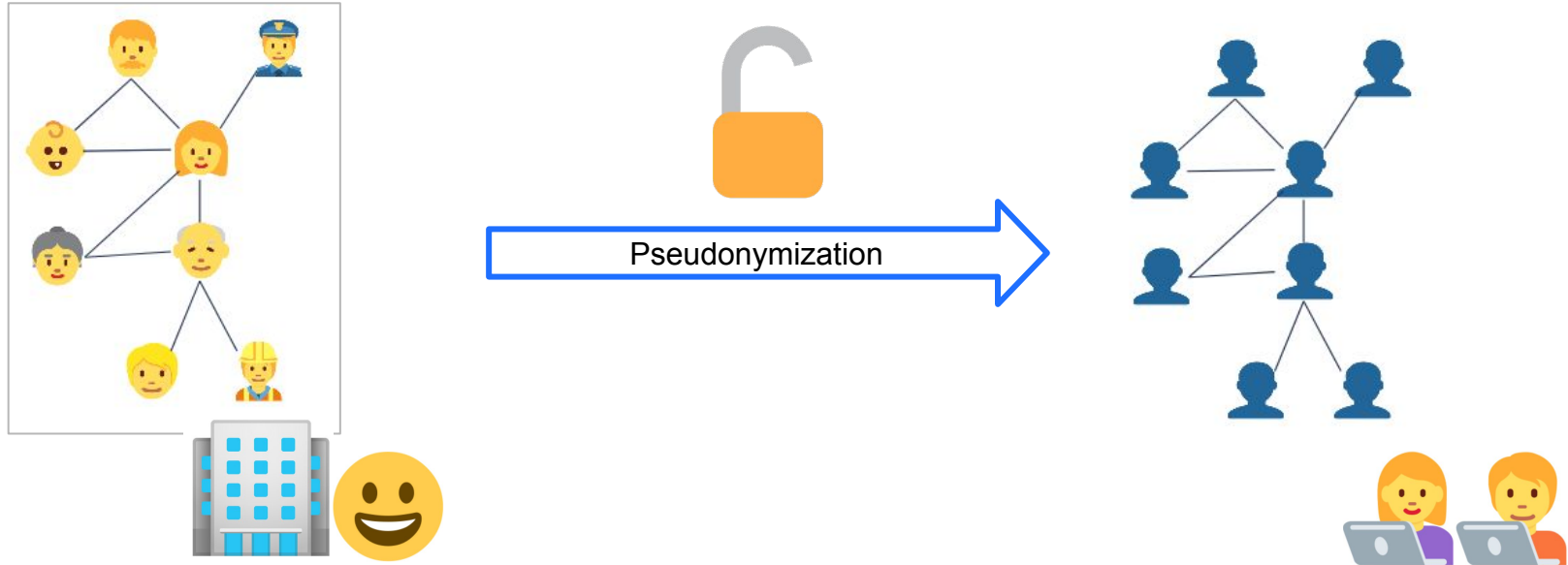
Universiteit
Leiden

The Anonymization Problem in Social Networks

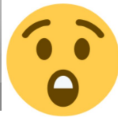
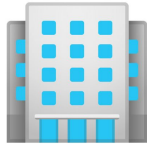
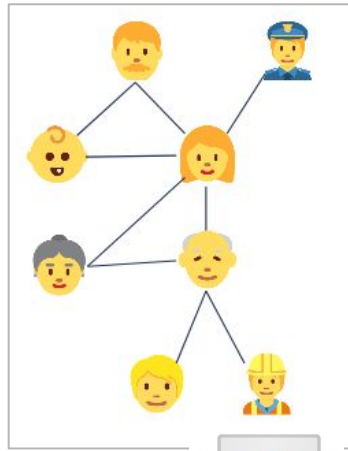
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Mark P. J. van der Loo,
Prof. Frank W. Takes

Leiden University, Statistics Netherlands

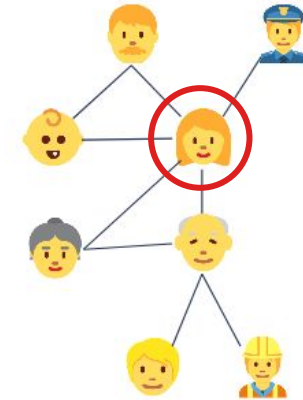
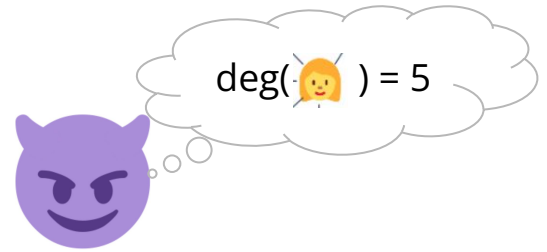
Safely publishing networks?



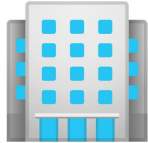
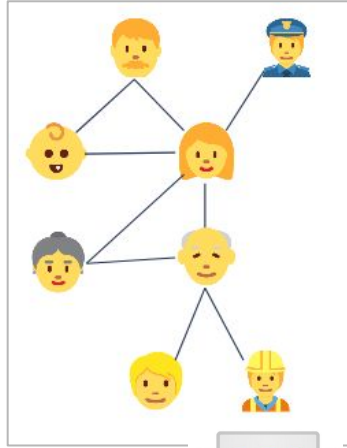
Safely publishing networks?



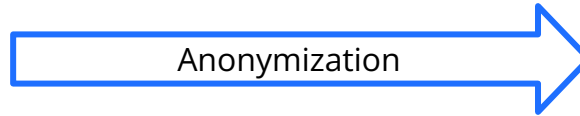
Pseudonymization



Safely publishing networks?

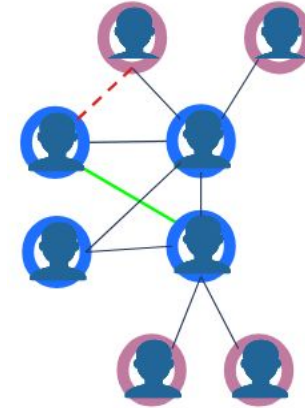


- How to measure **anonymity**?
- Preserving **utility**?
- **Anonymization**?



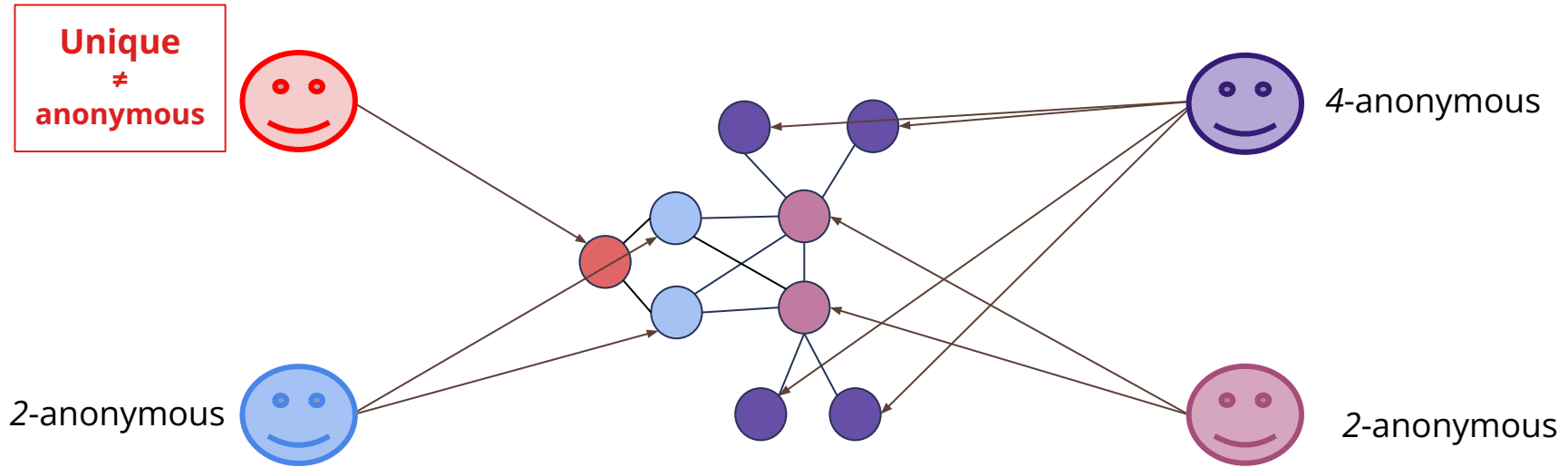
Deleting edges - - - - -

Adding edges - - - - -



k-Anonymity

How many candidates are there for each individual?



The network anonymization problem:

Given:

- **Network G ,**
- Anonymity **measure M**
- Set of allowed **alteration operations**

Goal:

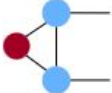
1. **Full:** Make **all** nodes *k-anonymous*
 - with *as few alterations* as possible.
2. **Partial:** Make **$\alpha\%$** of nodes *k-anonymous*
 - with *as few alterations* as possible.
3. **Budgeted:** make as many nodes as possible *k-anonymous*
 - With **B** edge alterations

The work so far

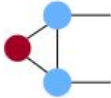
Full anonymization

Algorithms

- Measure agnostic:
 - Random
- Measure specific:
 - Degree (optimal)
 - Genetic algorithm
 - Make neighborhoods isomorphic

	$d = 1$	
	Reach	Value
Degree [31, 70–72]		2
Count [73]		(3, 3)
Degree distribution [73]		{2, 2, 2}
d - k -Anonymity [11, 12, 74, 75]		
VRQ [32, 76]		{2, 3, 3}
Hybrid [77]		 {2, 3, 3}

Our contribution

	$d = 1$	
	Reach	Value
Degree [31, 70–72]		2
Count [73]		(3, 3)
d - k -Anonymity [11, 12, 74, 75]		
VRQ [32, 76]		{2, 3, 3}

This work:

- Three variants
- 4 measures
(focus on COUNT)

Algorithms (5):

- Measure agnostic
- Heuristic
 - Random
 - Structure based
 - Uniqueness based

Why the measure matters



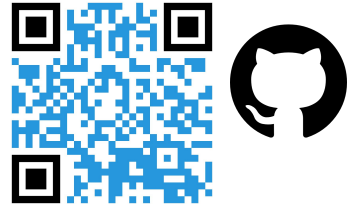


- Introduction
- **Approach**
- Results
- Conclusion

The ANO-NET package

C++ code to measure anonymity and perform anonymization with...

- Different anonymization **measures**
- Different settings + problem **variants**
- Different **anonymization algorithms**
 - (using edge deletion)



[github.com/RacheldeJong/
ANONET](https://github.com/RacheldeJong/ANONET)

Overall experimental setup

If I use a different...

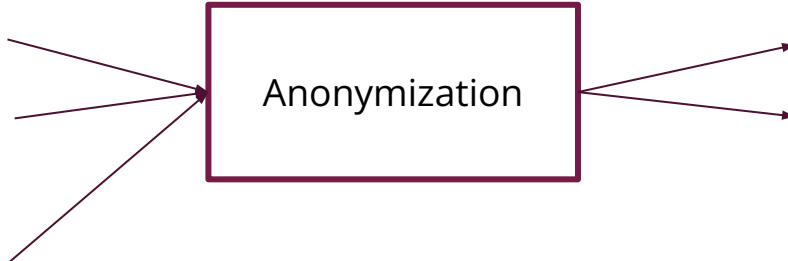
What happens to...

1. Alteration operation

2. Anonymity measure +
algorithm

3-4. Problem variant +
algorithm

Anonymization



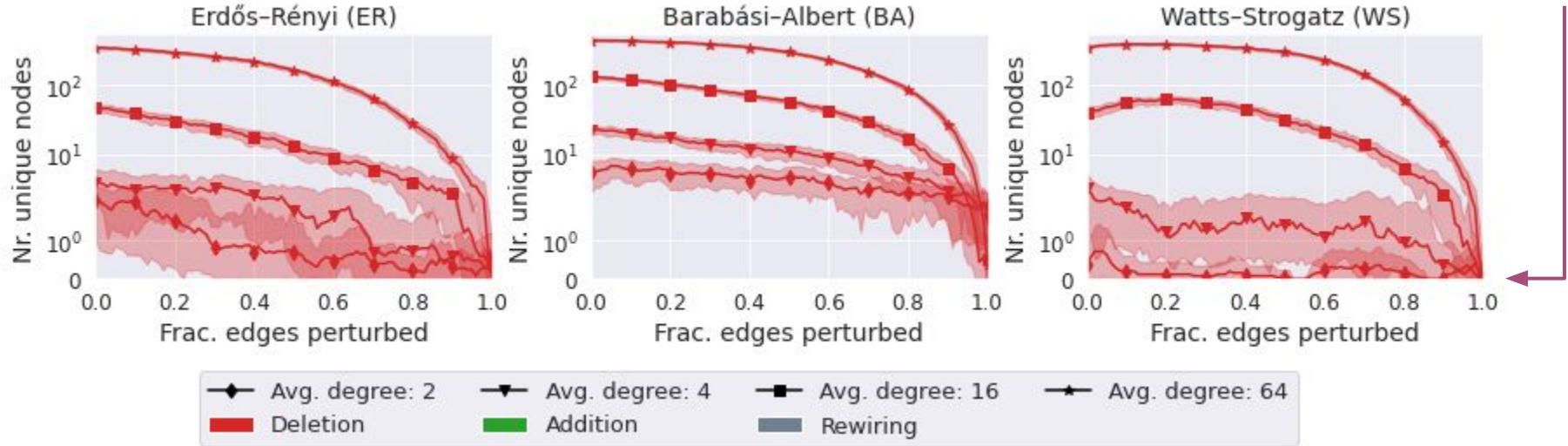
```
graph LR; A[1. Alteration operation] --> B[Anonymization]; C[2. Anonymity measure + algorithm] --> B; D[3-4. Problem variant + algorithm] --> B; B --> E[1-3. Uniqueness]; B --> F[4. Utility];
```

1-3. Uniqueness

4. Utility

Result 1: alteration operations

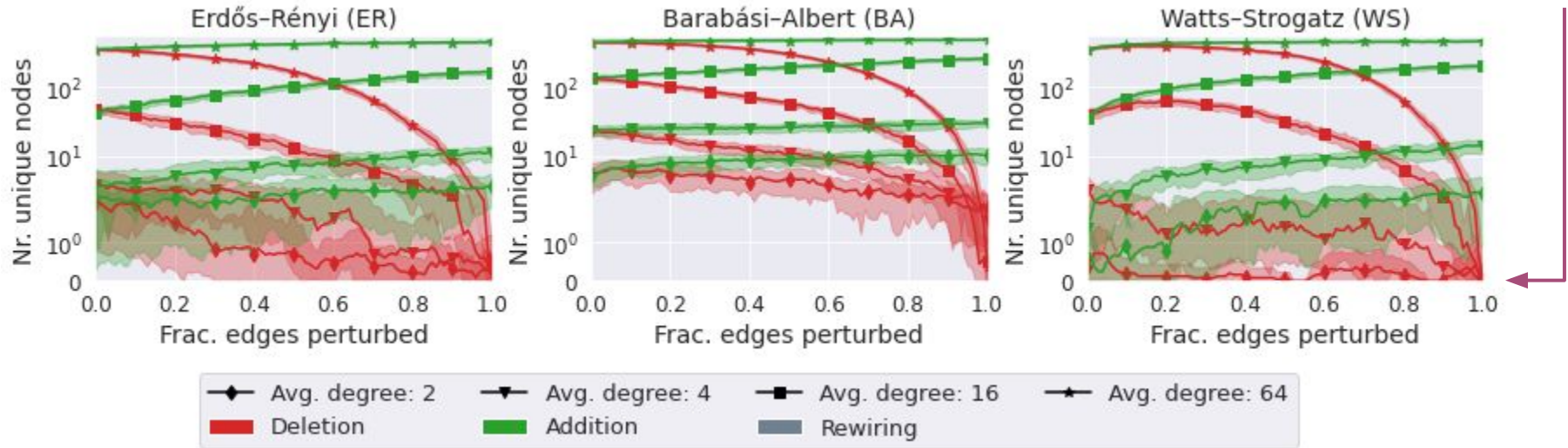
Lower is better!



- Higher density → higher uniqueness
- Deletion → decrease in anonymity
 - Towards an edgeless → graph, fewer possible "neighborhoods"

Result 1: alteration operations

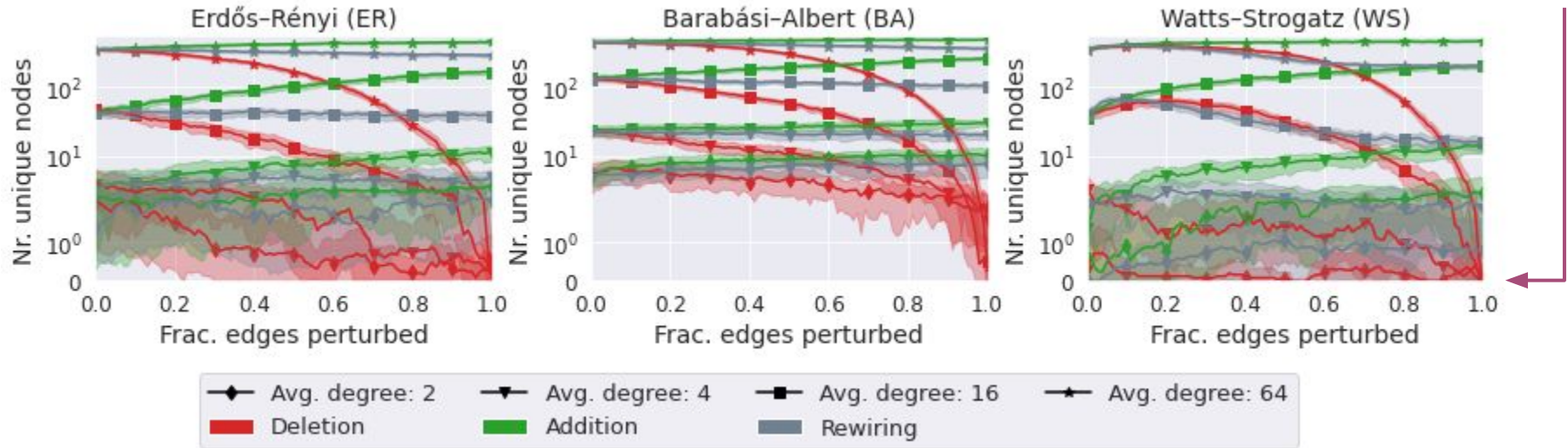
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- Higher density → higher uniqueness
- Deletion → decrease in anonymity
 - Towards an edgeless → graph, fewer possible "neighborhoods"
- Addition slight increase in anonymity

Result 1: alteration operations

Lower is better!



- **Higher density → higher uniqueness**
- **Deletion → decrease in anonymity**
 - Towards an edgeless → graph, fewer possible "neighborhoods"
- **Addition slight increase in anonymity**
- **Rewiring → towards ER graph**

Network datasets

		$ V $	$ E $	Avg. deg.	Clust. coeff	Frac. LCC	Avg. dist.	$U(G)$
Radoslaw emails	[10]	167	3,250	38.92	0.69	1.00	1.97	0.766
Primary school	[83]	242	8,317	68.74	0.53	1.00	1.73	0.975
Moreno innov.	[10]	241	923	7.66	0.31	0.49	2.47	0.245
Gene fusion	[10]	291	279	1.92	0.00	0.08	3.90	0.024
Copnet calls	[84]	536	621	2.32	0.25	0.65	7.37	0.024
Copnet sms	[84]	568	697	2.45	0.22	0.80	7.32	0.026
Copnet FB	[84]	800	6,418	16.05	0.32	1.00	2.98	0.488
FB Reed98	[18]	962	18,812	39.11	0.33	1.00	2.46	0.778
Arenas email	[10]	1,133	5,451	9.62	0.25	1.00	3.61	0.230
Euroroads	[10]	1,174	1,417	2.41	0.02	0.03	18.37	0.003
Air traffic control	[10]	1,226	2,408	3.93	0.07	1.00	5.93	0.042
Network science	[10]	1,461	2,742	3.75	0.88	0.00	5.82	0.039
FB Simmons81	[18]	1,518	32,988	43.46	0.33	0.99	2.57	0.785
DNC emails	[10]	1,866	4,384	4.70	0.59	0.98	3.37	0.092
Moreno health	[10]	2,539	10,455	8.24	0.15	1.00	4.56	0.054
US power grid	[10]	3,783	14,124	7.47	0.28	1.00	3.57	0.008
Bitcoin alpha	[18]	4,941	6,594	2.67	0.11	1.00	18.99	0.113
GRQC collab.	[17]	5,241	14,484	5.53	0.69	0.79	6.05	0.054
Pajek Erdős	[10]	6,927	11,850	3.42	0.40	1.00	3.78	0.044
FB GWU54	[18]	12,193	469,528	77.02	0.22	1.00	2.83	0.682
Enron email	[17]	36,692	183,831	10.02	0.72	0.92	4.03	0.071
FB wall 2009	[10]	45,813	183,412	8.01	0.15	0.96	5.60	0.033
Brightkite	[18]	58,228	214,078	7.35	0.27	0.97	4.92	0.048
Twitter	[10]	465,017	833,540	3.59	0.06	1.00	4.59	0.004

Random and Structure based anonymization

1. Edge Sampling (ES)

- Random

2. DEGREE:

- Target edges connecting high degree nodes

$$P_{\text{DEGREE}}(\{v, w\}) = \frac{\min(\text{degree}(v), \text{degree}(w))}{\sum_{\{v', w'\} \in E} \min(\text{degree}(v'), \text{degree}(w'))}$$

Random and Structure based anonymization

1. Edge Sampling (ES)

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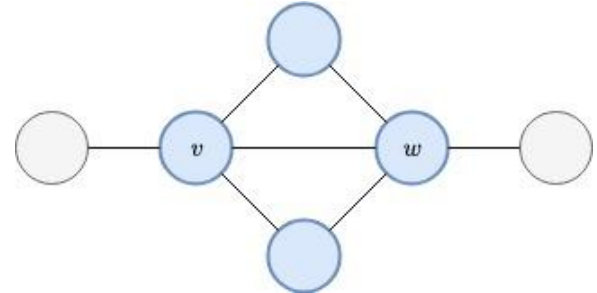
$$P_{\text{DEGREE}}(\{v, w\}) = \frac{\min(\text{degree}(v), \text{degree}(w))}{\sum_{\{v', w'\} \in E} \min(\text{degree}(v'), \text{degree}(w'))}$$

3. Affected (AFF):

- Target edges affecting many nodes

$$P_{\text{AFF}}(\{v, w\}) = \frac{|A_M(\{v, w\})|}{\sum_{\{v', w'\} \in E} |A_M(\{v', w'\})|}$$

$A_M = v, w$ and nodes connected to v and w



Uniqueness based anonymization

4. UNIQUE

- Delete edges connected to unique node(s)

$$P_{\text{UNIQUE}}(\{v, w\}) = \begin{cases} 1/|E_u| & \text{if } \{v, w\} \in E_u, |E_u| > B' \\ 1 & \text{if } \{v, w\} \in E_u, |E_u| \leq B' \\ 1/|E \setminus E_u| & \text{if } \{v, w\} \notin E_u, |E_u| < B' \\ 0 & \text{otherwise} \end{cases}$$

Uniqueness based anonymization

4. UNIQUE

- Delete edges connected to unique node(s)

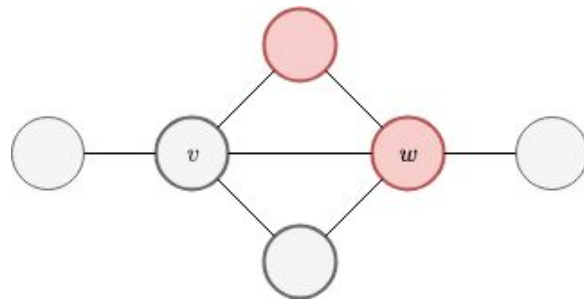
$$P_{\text{UNIQUE}}(\{v, w\}) = \begin{cases} 1/|E_u| & \text{if } \{v, w\} \in E_u, |E_u| > B' \\ 1 & \text{if } \{v, w\} \in E_u, |E_u| \leq B' \\ 1/|E \setminus E_u| & \text{if } \{v, w\} \notin E_u, |E_u| < B' \\ 0 & \text{otherwise} \end{cases}$$

5. Unique Affected (UA)

- Delete edges **affecting** more **unique nodes**

$$P_{\text{UA}}(\{v, w\}) = \frac{|A_M(\{v, w\}) \cap V_u| + 1/|E|}{\sum_{\{v', w'\} \in E} (|A_M(\{v', w'\}) \cap V_u| + 1/|E|)}$$

In A_M and unique





- Introduction
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Overall experimental setup

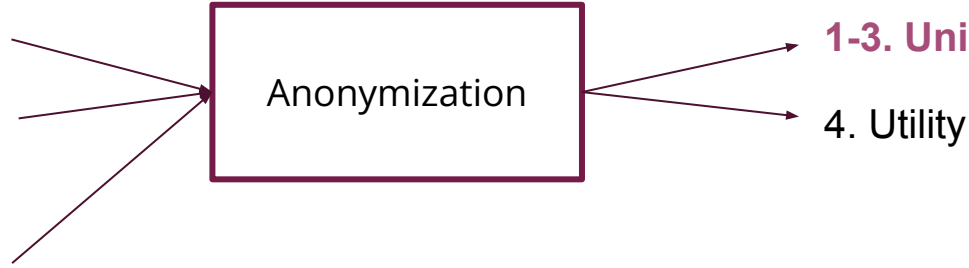
If I use a different...

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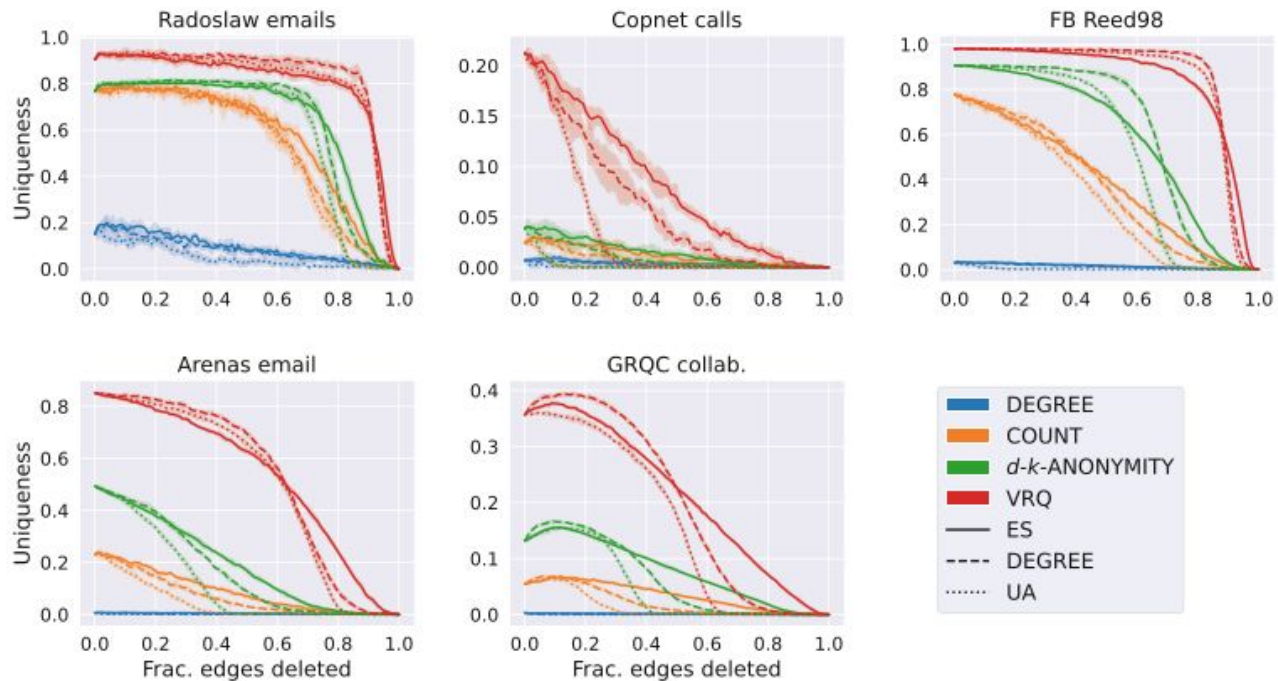
1. Alteration operation

**2. Anonymity measure
+ algorithm**

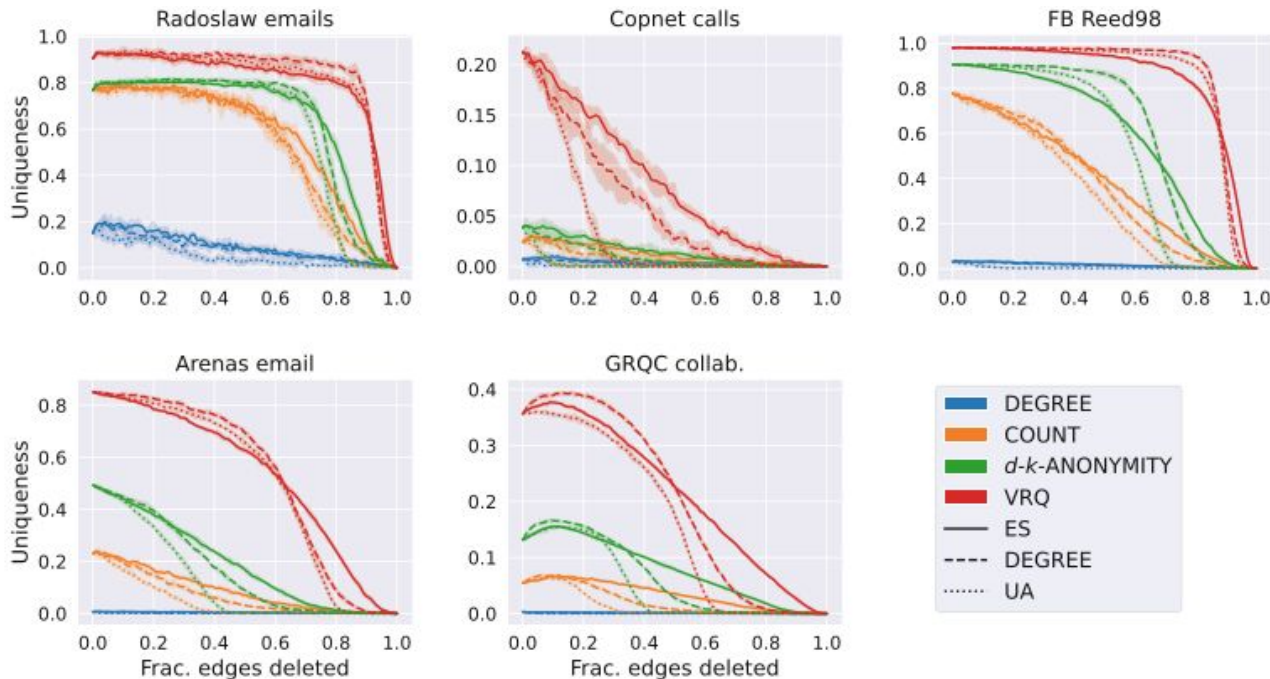
3-4. Problem variant +
algorithm



Result 2: measures



Result 2: measures

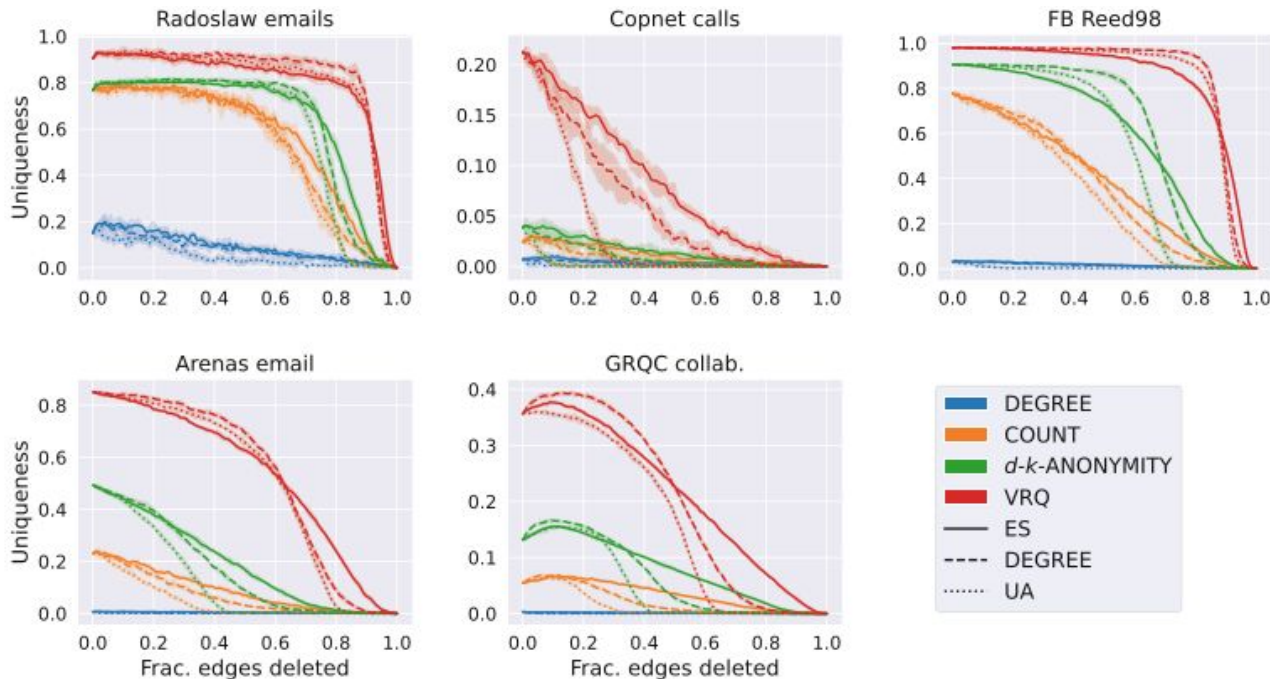


Lower is better!

- Measure: higher uniqueness → difficult anonymization → more edge deletions
- VRQ substantially more difficult

→ **focus on COUNT!**

Result 2: measures



Lower is better!

- Measure: higher uniqueness → difficult anonymization → more edge deletions
- VRQ substantially more difficult
- **focus on COUNT!**
- Algs → improvement
 - Unique Affected best

Overall experimental setup

If I use a different...

What happens to...

1. Alteration operation

2. Anonymity measure +
algorithm

**3-4. Problem variant +
algorithm**

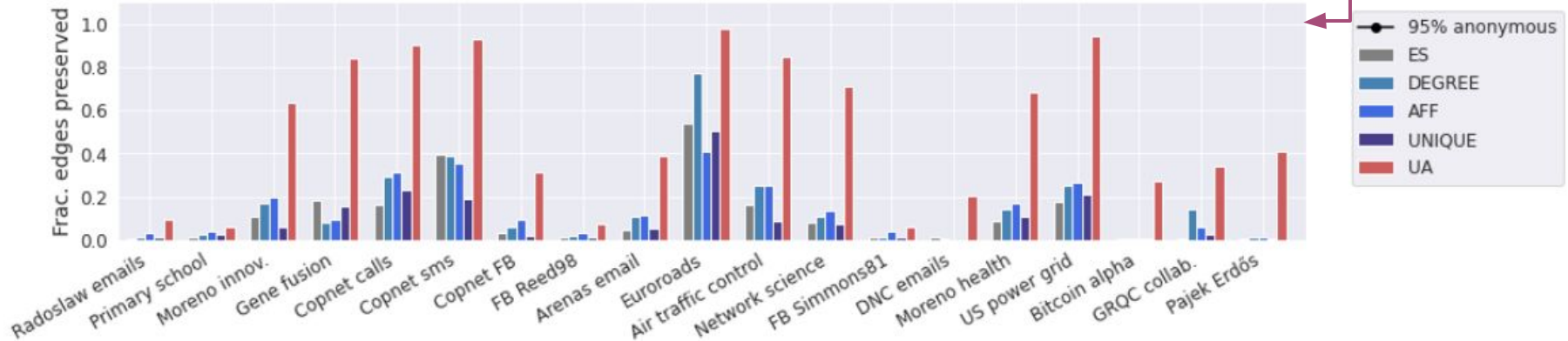


1-3. Uniqueness

4. Utility

Result 3: full anonymization

Higher is better!



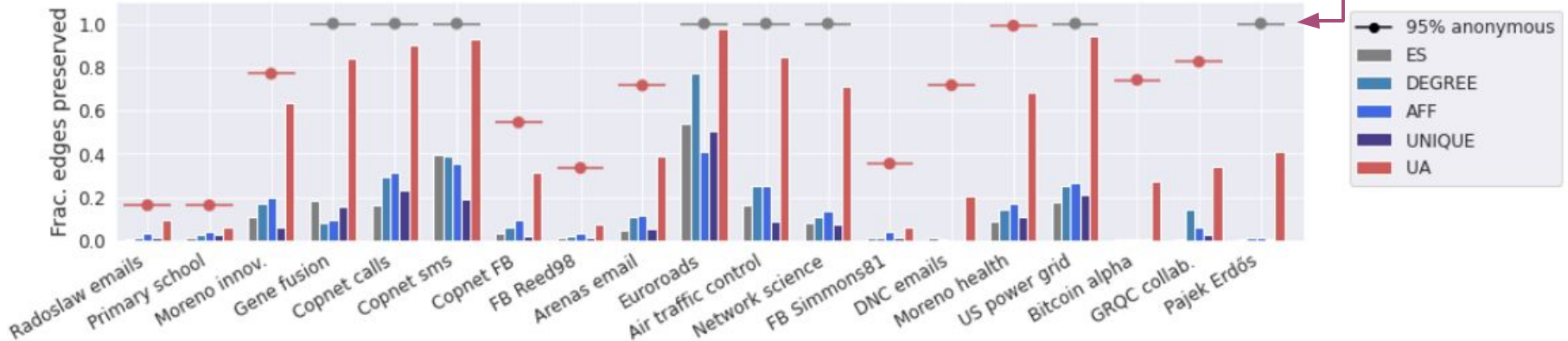
UA best: avg. 13.9 times more edges than ES

Works better on networks with

- Low initial **uniqueness**
- Lower average **degree**
- Higher average **distance**

Result 3: full and partial anonymization

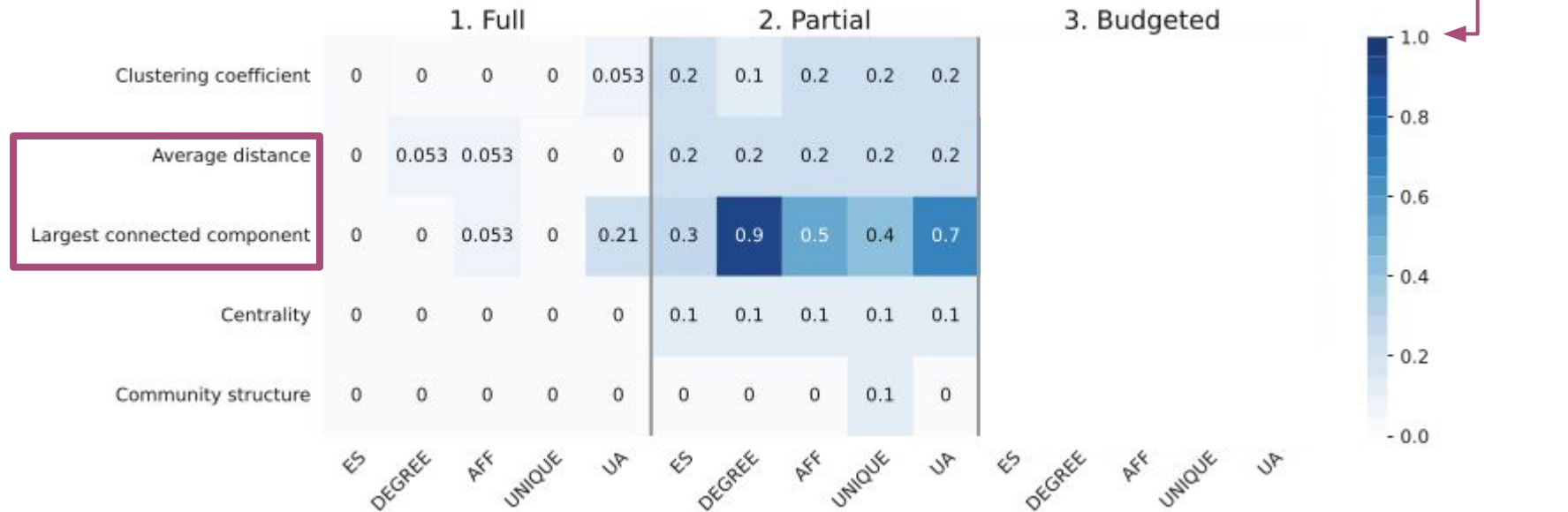
Higher is better!



UA best:

- avg. 1.8 times more edges than ES
- (some networks already partially anonymous)

Result 4: utility

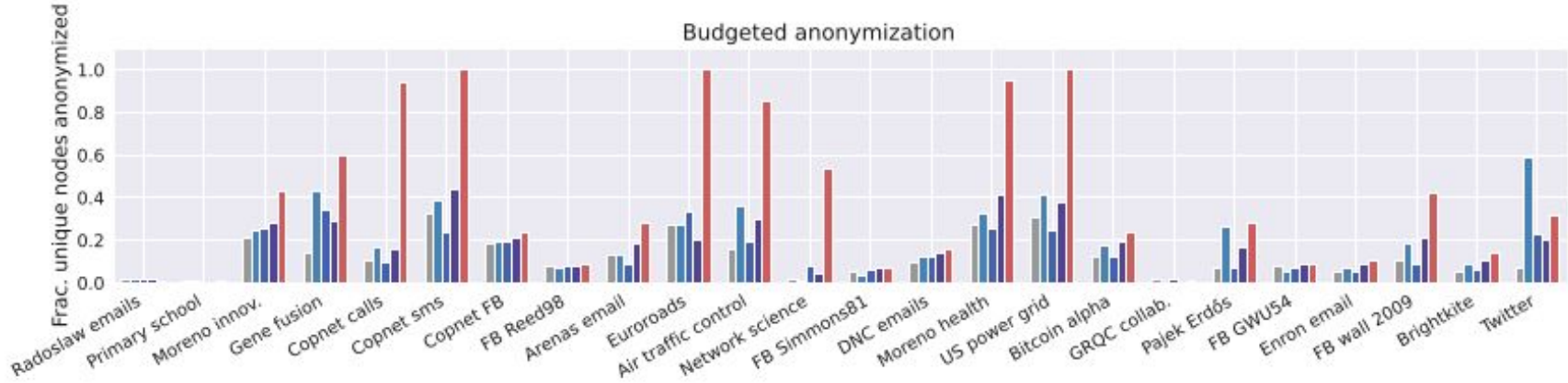


Frac. networks utility metric "preserved"

- Change + stdev $\leq 5\%$
- Average distance, largest connected component *easier* to preserve

Result 3: budgeted anonymization

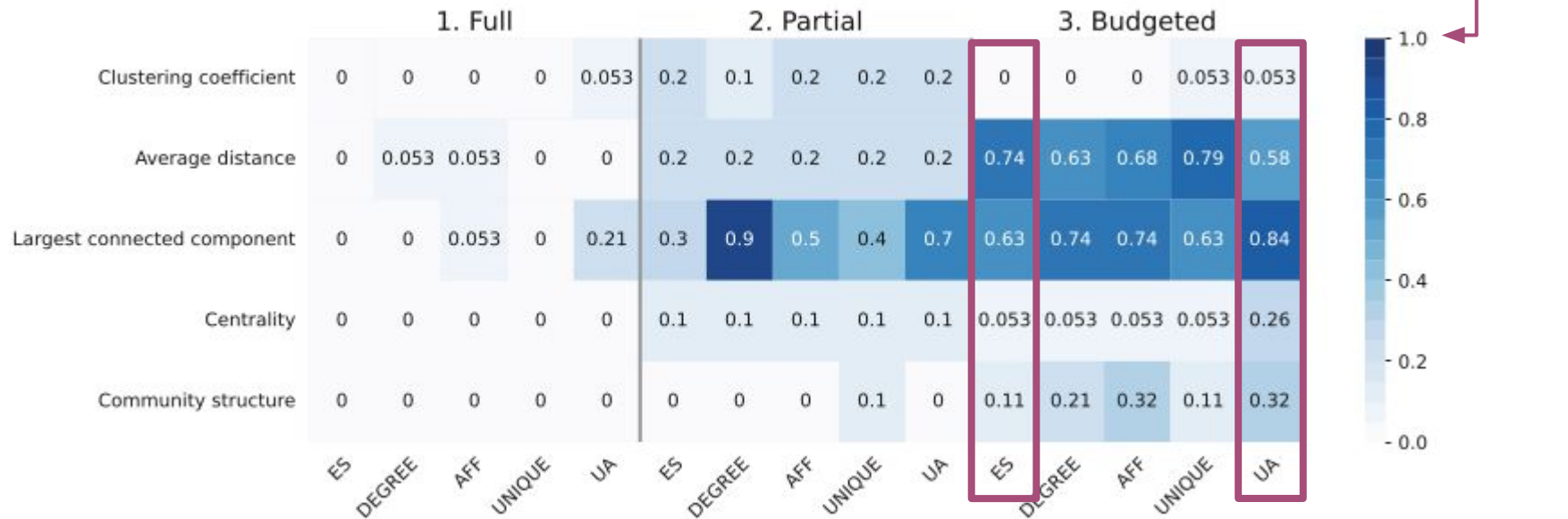
Higher is better!



UA avg. 4.8 times more anonymized than ES

- Fully anonymous for some networks
- (not all)

Result 4: utility



- **Budgeted:** more often properties preserved
- Random (ES) not always preserved
- UA (targeted) preserves better than ES

Thank you!

Summary

- Study the anonymization problem and its aspects
- Edge **deletion** more effective than addition / rewiring
 - Sparse networks → more anonymous nodes
- **Strict measure** → **higher uniqueness**, anonymization more challenging
- Better **heuristic algorithm** → improvement over baseline
 - Alg. targeting large nr. unique nodes most effective
- **Budgeted anonymization** promising if utility is important

Future work

- More effective anonymization algorithms
- Utility-aware anonymization
- *Fuzzy k -anonymity*
- Much more!

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Website

<https://racheldejong.github.io/>

ANO-NET package

github.com/RagheldeJong/ANONET

